

Term Deposit Subscription Prediction

Final Technical Report

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Executive Summary

A European bank wants to improve the success of telephone campaigns for term deposits. The practical decision is not whether a customer is "good" or "bad"; it is **which customers should be contacted first when call capacity is limited**. This project uses 40,000 historical calls to compare one parametric model, Logistic Regression, with one tree-based model, HistGradientBoosting.

The analysis follows the project requirements in a controlled sequence: Exploratory Data Analysis (EDA), baseline models with and without call duration, hyperparameter tuning, feature reduction, class-imbalance methods, target-customer modelling, and a final robustness check on later campaign periods.

The main development result is that HistGradientBoosting provides the strongest shuffled cross-validation ranking performance using pre-call features (ROC-AUC about 0.72 and PR-AUC about 0.25). However, expanding-window validation on later campaign periods reduces both models substantially, and Logistic Regression is more stable forward in time. The headline baseline and rolling results were also reproduced with a separately written verification pipeline; the central ROC-AUC values agree within approximately 0.01-0.02. Therefore:

- **duration** is excluded from all usable models because it is only known after the call;
- HistGradientBoosting is the strongest **development** model on randomly shuffled folds;
- unweighted Logistic Regression using all 12 pre-call features is the safer **pilot** model because it transfers better to future periods and is easier to explain;
- no class-imbalance method improves ranking; they mainly trade precision for recall;
- the EDA-derived target population has a higher subscription rate, but dedicated segment models do not outperform the full-data approach reliably enough to justify a separate production model.

A **pilot** means a limited real campaign test in which model-ranked calling is compared with current practice before any bank-wide rollout.

1. Business Problem and Project Objectives

1.1 Business problem

Every marketing call costs staff time. Calling all eligible customers maximises reach but also creates many unproductive calls. A useful model should rank customers so that, at a fixed call budget, more subscribers are contacted earlier.

The target variable is y :

- **yes**: the customer subscribed to the term deposit;
- **no**: the customer did not subscribe.

The model is intended for **pre-call prioritisation**, so only information available before the call may be used.

1.2 Project aims

The project answers eight concrete questions:

1. What does the dataset contain, and which patterns are visible in EDA?
2. Why were Logistic Regression and HistGradientBoosting selected?
3. How do both models perform as untuned baselines?
4. How much performance is caused by the unavailable **duration** feature?
5. Do systematic hyperparameter searches improve either model?
6. Can the tuned models retain performance using only the top 3-5 features?
7. Which class-imbalance method works best for the parametric and tree-based models?
8. Can EDA identify a useful target population, and do dedicated models improve on the full-data models?

A final expanding-window analysis checks whether the development conclusions remain valid on later campaign periods.

2. Dataset and Evaluation Framework

2.1 Dataset

Each row is one completed marketing call. Table 1 summarises the dataset size, target prevalence and data-quality indicators that define the modelling problem.

Table 1. Dataset composition and data-quality summary.

Property	Value
Calls	40,000
Subscriptions	2,896
Subscription rate	7.24%
Non-subscriptions	37,104
Conventional null values	0
Explicit unknown categories	Present in contact, education, and job

2.2 Development validation

The main modelling experiments use **stratified five-fold cross-validation**. The data is split into five folds while preserving the 7.24% class ratio. Each model trains on four folds and predicts the fifth; this repeats five times so every customer receives an out-of-fold prediction.

All preprocessing happens inside each training fold:

- numeric variables are standardised;
- categorical variables are one-hot encoded;
- oversampling, undersampling and SMOTENC are applied only to training folds;
- the held-out fold is never used to fit preprocessing or sampling steps.

2.3 Metrics and confusion matrices

Because only 7.24% of customers subscribe, accuracy alone is misleading: an always-no model already reaches 92.76%. Table 2 therefore defines the ranking and threshold metrics used together throughout the report.

Table 2. Evaluation metrics used for the imbalanced target.

Measure	Beginner-friendly meaning
Accuracy	Percentage of all predictions that are correct
Precision	Of customers predicted to subscribe, how many actually subscribe?
Recall	Of all real subscribers, how many did the model find?
F1	Balance between precision and recall
ROC-AUC	How well the model ranks subscribers above non-subscribers across all thresholds; 0.5 is random
PR-AUC	Ranking quality focused on the rare subscriber class; the no-skill reference is 0.0724

A confusion matrix reports four counts: true negatives (TN), false positives (FP), false negatives (FN) and true positives (TP). These counts make the practical trade-off visible: increasing recall usually creates more false-positive calls.

3. Exploratory Data Analysis (EDA)

EDA was used to understand data quality, class balance, distributions, correlations, campaign structure and possible target populations before modelling.

Figure 1. Target distribution: only 7.24% of customers subscribed.

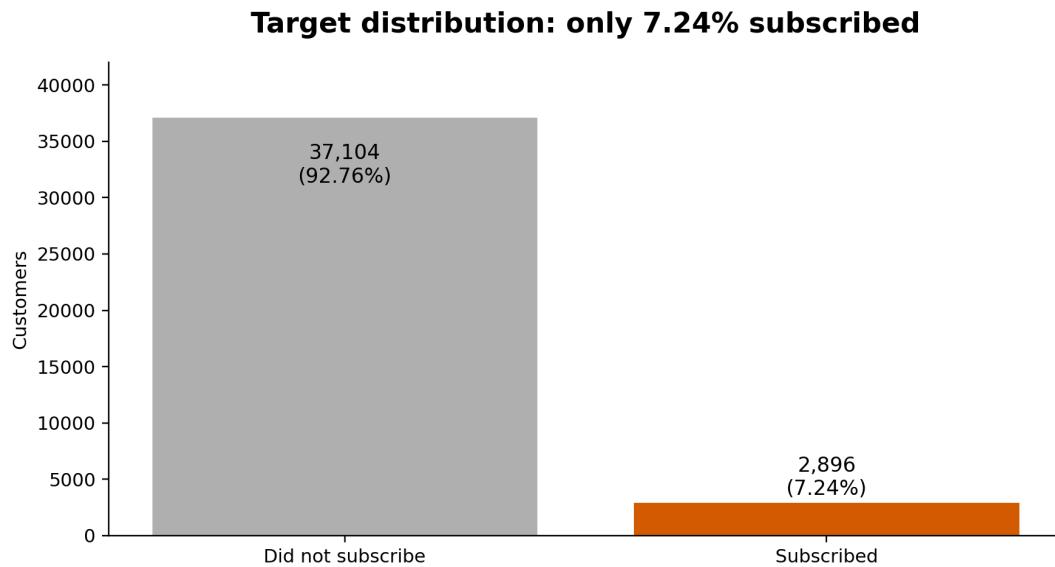


Figure 1 shows that only 2,896 of 40,000 calls (7.24%) produced a subscription. The severe imbalance means accuracy can look high even when subscribers are rarely identified, so recall, F1, ROC-AUC and PR-AUC are reported alongside it.

Figure 2. Explicit unknown categories, led by contact at 31.9% of records.

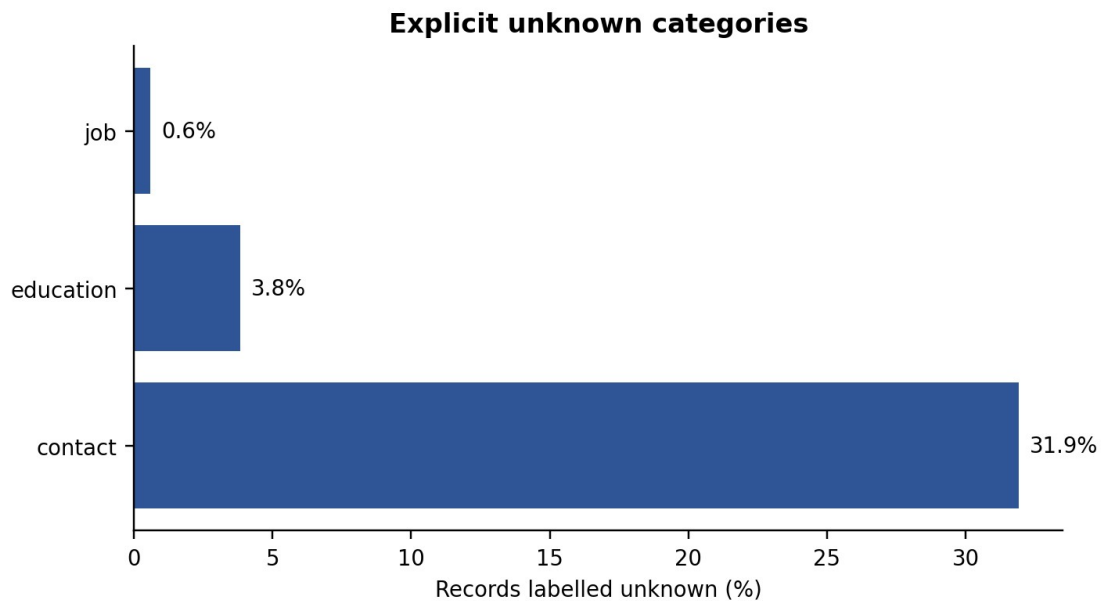


Figure 2 shows that contact=unknown is the dominant missing-information category, affecting about 31.9% of records. These rows cannot simply be deleted because they include genuine subscribers and may represent an operational data-quality problem.

Figure 3. File-order structure: contiguous month blocks and changing response rates.

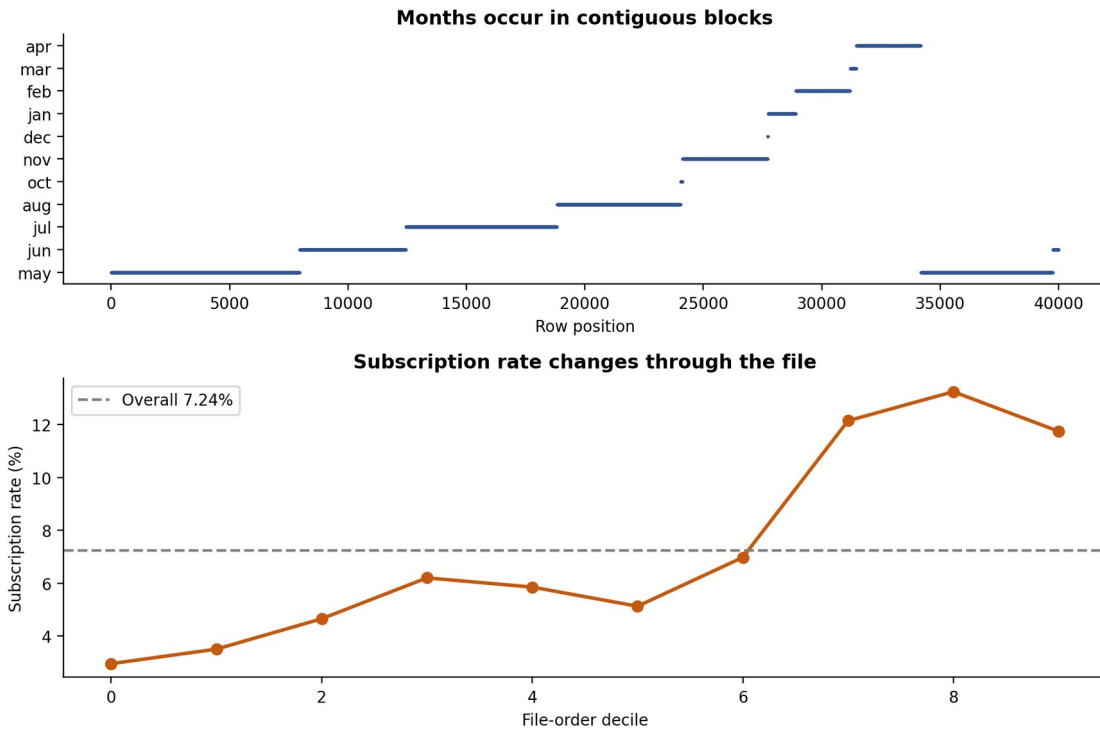


Figure 3 shows that month labels occur in contiguous file-order blocks while subscription rates change across the same sequence. This ordered structure motivates the later expanding-window robustness check because campaign conditions are not constant.

Figure 4. Subscription rate by file-order decile, showing campaign-period drift.

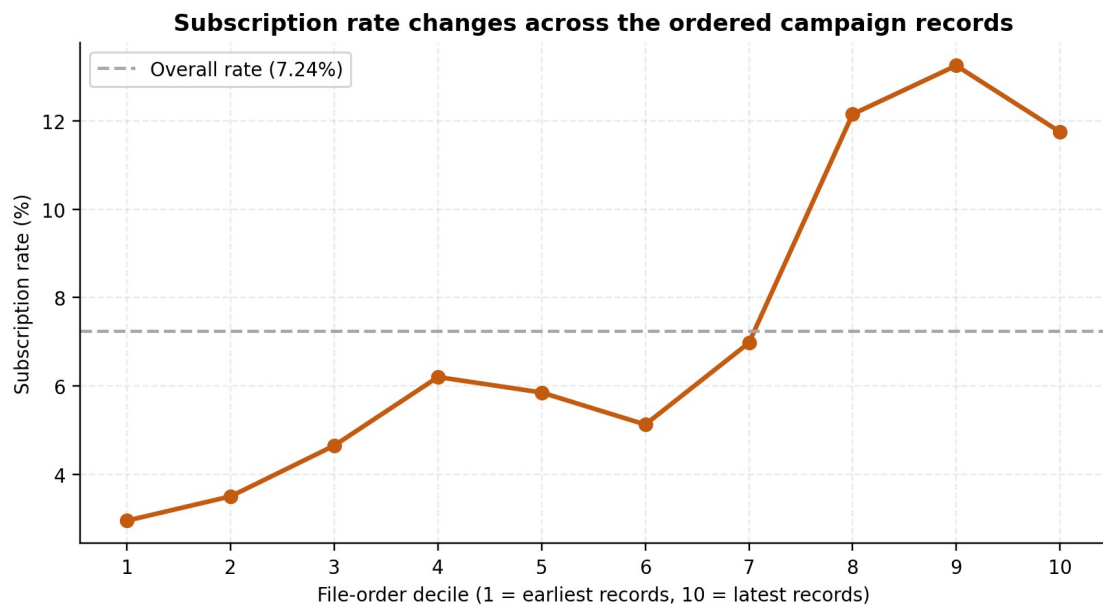


Figure 4 isolates the same pattern by file-order decile. The rise and fall in response rates is descriptive evidence of campaign-period drift and shows why a random split can mix easier and harder periods.

Figure 5. Conversion-rate patterns across month, contact, job and housing groups.

EDA: conversion rates by category (grey = fewer than 300 records)

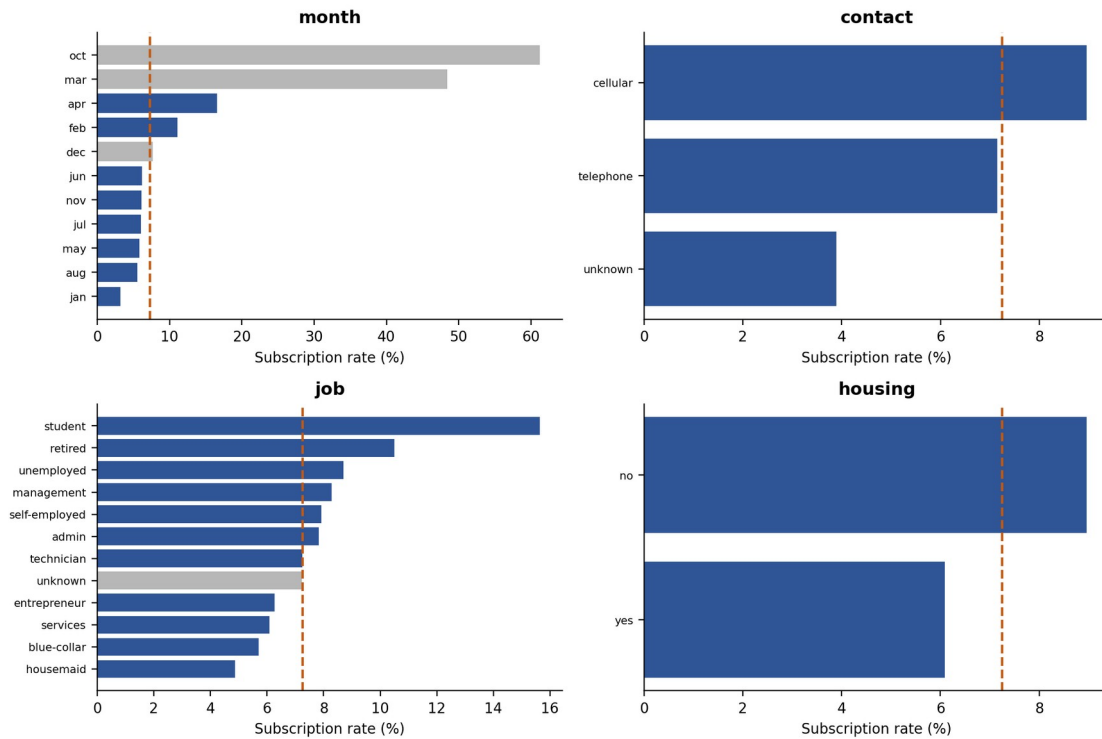
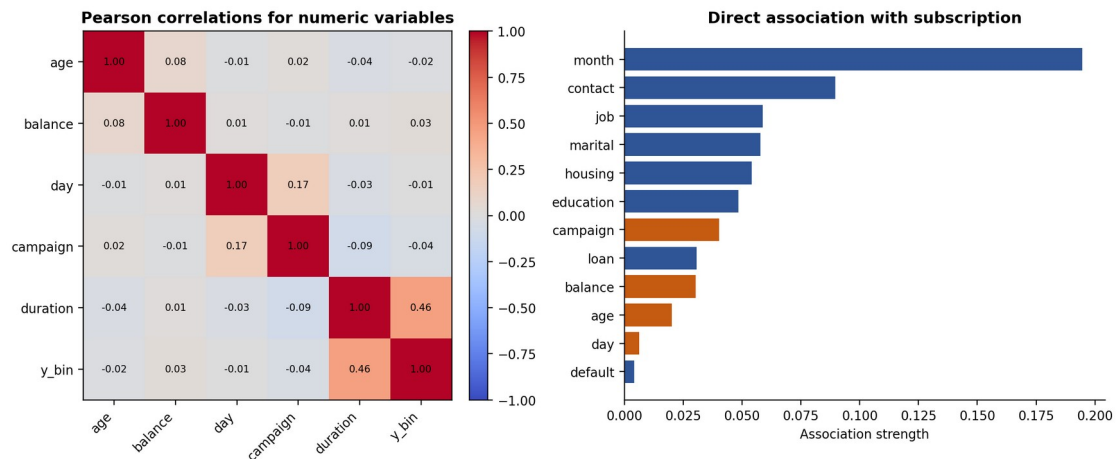


Figure 5 compares raw conversion rates across month, contact, job and housing groups. Month and contact show the clearest differences, while retired, older and high-balance cellular customers provide the exploratory signal used later in Section 9. Small groups are not treated as reliable merely because their observed rate is high.

3.1 Correlations and associations

Pearson correlation was used for numeric variables (-1 to +1), while Cramer's V measured categorical association strength from 0 upward. Figure 6 places these complementary measures side by side so direct numeric and categorical relationships can be compared without treating them as the same statistic.

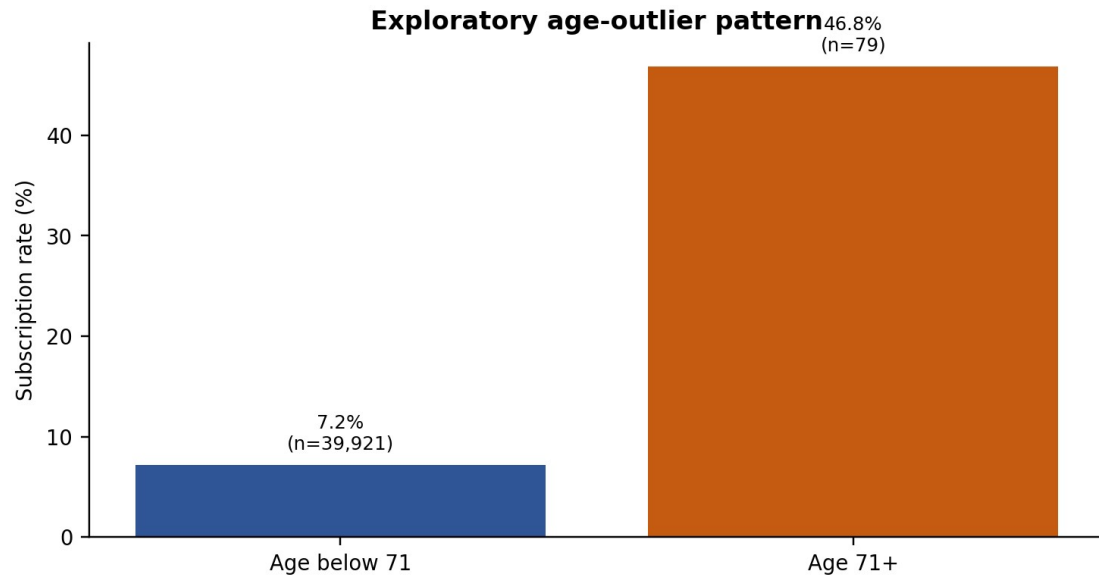
Figure 6. Direct associations with subscription: Pearson r and Cramer's V.



Key findings:

- **duration** has the largest numeric correlation with the target ($r = 0.461$), but it is unavailable before the call;
- no two usable numeric predictors are strongly correlated; the largest is **day** with **campaign** ($r = 0.165$);
- **month** has the strongest categorical association with subscription (Cramer's V about 0.195);
- **contact**, **job**, **marital**, **housing** and **education** have weaker but non-zero associations;
- weak individual correlation does not imply a feature is useless, because tree models can use interactions and non-linear patterns.

EDA therefore suggested useful campaign patterns, but not a single stable pre-call variable capable of explaining the outcome on its own.

Figure 7. Exploratory age split: observed subscription rate for age ≥ 71 .

The 71+ group is small and data-derived; this is a lead for validation, not a targeting rule.

Figure 7 shows the exploratory age split: customers aged 71 or older have a high raw subscription rate, but the group contains only 79 customers and the threshold was discovered from the same data. It is therefore a lead for monitoring and fairness analysis, not an automatic targeting rule.

4. Preprocessing, Model Choice and Experimental Design

4.1 Why Logistic Regression?

Logistic Regression is the parametric model. It estimates subscription probability from a weighted linear combination of the encoded features. It was chosen because it is:

- a standard classification baseline;
- transparent and easy to explain;
- fast to tune and deploy;
- less likely than a complex model to learn unstable campaign-specific interactions.

4.2 Why HistGradientBoosting?

HistGradientBoosting is the tree-based model. It combines many small decision trees and can learn:

- non-linear relationships;
- interactions between variables;
- different effects in different regions of the data.

It was chosen to test whether added flexibility improves performance enough to justify greater complexity.

4.3 Controlled experiment matrix

The modelling sequence isolates one question at a time:

1. untuned baseline models with all variables;
2. the same models after removing **duration**;
3. tuned versions of all four configurations;
4. tuned no-duration models transferred to top-3, top-4 and top-5 features;
5. six imbalance strategies for each model family;
6. full-data models versus EDA-derived target-population models;
7. expanding-window validation as a robustness check.

5. Baseline Models and Leakage Investigation

5.1 What is leakage?

Leakage occurs when a model is given information that would not exist when the real prediction is made. **duration** is the final length of the call. Before calling a customer, the final call length is unknown. Therefore **duration** may be analysed diagnostically, but it cannot be used for pre-call targeting.

The leakage test compares the same model family with and without duration. Table 3 reports baseline and tuned performance on the untouched 20% final holdout (8,000 rows), so the tuning set does not also grade the selected model.

Stage	Model	Features	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Baseline	Logistic Regression	Pre-call	0.928	0.576	0.033	0.062	0.711	0.242
Baseline	Logistic Regression	With duration	0.938	0.656	0.306	0.417	0.932	0.525
Baseline	HistGradientBoosting	Pre-call	0.928	0.535	0.040	0.074	0.734	0.268
Baseline	HistGradientBoosting	With duration	0.941	0.640	0.427	0.512	0.952	0.585
Tuned	Logistic Regression	Pre-call	0.928	0.559	0.033	0.062	0.710	0.240
Tuned	Logistic Regression	With duration	0.938	0.654	0.290	0.402	0.933	0.523
Tuned	HistGradientBoosting	Pre-call	0.927	0.474	0.016	0.030	0.730	0.270
Tuned	HistGradientBoosting	With duration	0.941	0.658	0.396	0.494	0.951	0.581

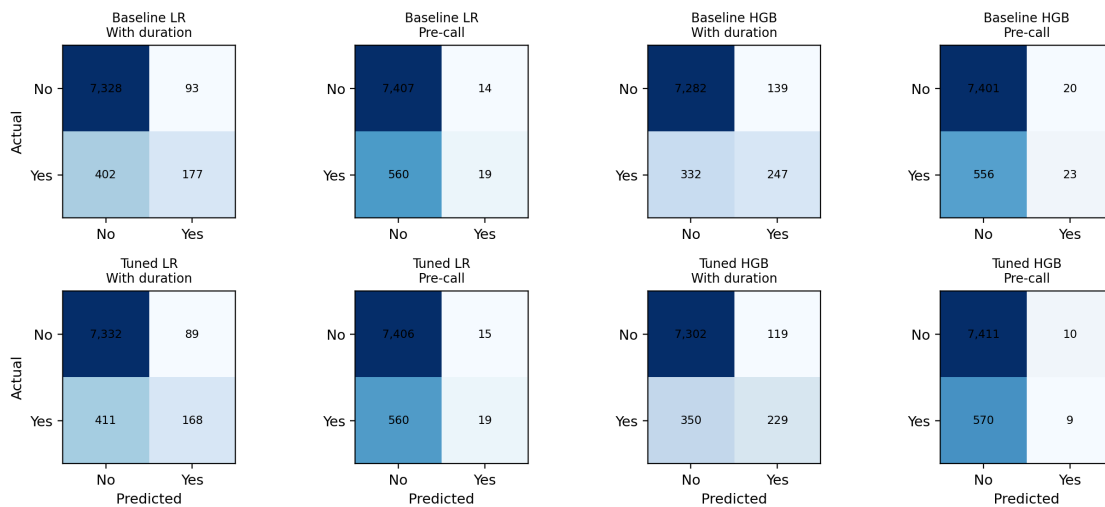


Figure 9. ROC-AUC with versus without post-call duration.

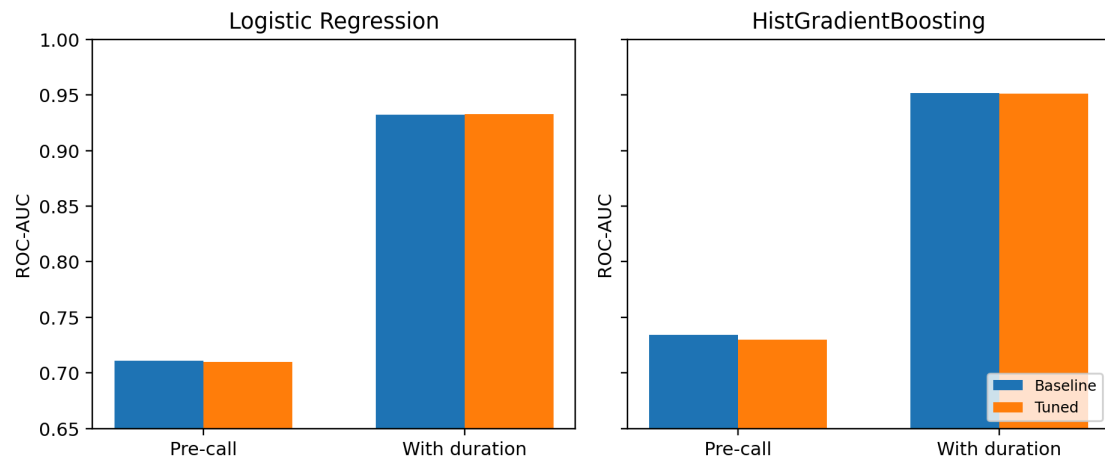


Figure 9 shows the approximately 0.22-0.24 ROC-AUC increase produced by duration, while Figure 8 on the preceding page shows the corresponding classification behaviour. The gain is diagnostic evidence of post-call leakage, not a deployable improvement.

Conclusion: all final targeting models exclude duration.

6. Hyperparameter Tuning

Hyperparameters are settings chosen before training. Tuning was performed to determine whether better model settings could improve PR-AUC on the imbalanced target.

6.1 Search design

Parameter selection was separated from final evaluation. The 80% development set was split into tuning-train and tuning-validation data; the final 20% holdout remained untouched until the selected configurations were evaluated. PR-AUC was the selection metric.

Model	Hyperparameter	Range	Why it was searched
Logistic Regression	C	15 log-spaced values from 0.001 to 100	Tests strong to weak regularisation
	penalty	L1, L2	L1 can remove redundant encoded variables; L2 shrinks all coefficients
	class weight	none, balanced	Tests whether the 7.24% minority class needs extra weight
HistGradientBoosting	iterations	50, 100, 150, 200, 300, 400	Controls how many boosting stages are added
	maximum depth	3, 4, 5, 6, 8, unlimited	Controls interaction complexity
	learning rate	0.01, 0.03, 0.05, 0.1, 0.2	Controls how strongly each tree updates the model
	leaf nodes	15, 31, 63, 127	Controls tree complexity
	L2 regularisation	0, 0.1, 0.5, 1, 2	Penalises overly complex fits
	minimum samples per leaf	10, 20, 30, 50	Prevents tiny unstable leaves
	class weight	none, balanced	Tests minority-class emphasis

Logistic Regression has 60 unique combinations and all 60 were evaluated for each feature set. HistGradientBoosting has 28,800 possible combinations; 100 reproducible random candidates were evaluated for each feature set using a stratified 10,000-row tuning subsample and the separate tuning-validation set. Across the four searches, 320 candidate configurations were run.

6.2 Selected parameters

Configuration	Selected parameters
LR with duration	L2, C=0.139, unweighted
LR pre-call	L2, C=8.483, unweighted
HGB with duration	200 iter, depth 8, lr 0.05, 15 leaves, L2=0, min leaf 20, unweighted
HGB pre-call	200 iter, depth 8, lr 0.05, 15 leaves, L2=0, min leaf 20, unweighted

Table 5. Selected hyperparameters for the four model configurations.

Table 5 summarises the selected settings. All four selected configurations are unweighted. On the untouched holdout, tuning produced little or no ranking improvement over the defaults, so the later analysis treats tuning as a model-selection exercise rather than evidence that extra complexity automatically improves generalisation.

7. Feature Importance and Reduced Models

7.1 How importance was investigated

Feature importance was not inferred from correlation alone. Three complementary views were used:

1. **Pearson correlation and Cramer's V:** direct one-feature association with the target;
2. **model-based contribution:** the change in prediction when a feature is removed or permuted;
3. **partial dependence:** how the fitted model's average prediction changes across values of an important feature.

These methods answer different questions. Correlation shows direct association; model importance shows whether a feature contributes after other variables are present.

Figure 10. Model-based feature importance for Logistic Regression and HGB.

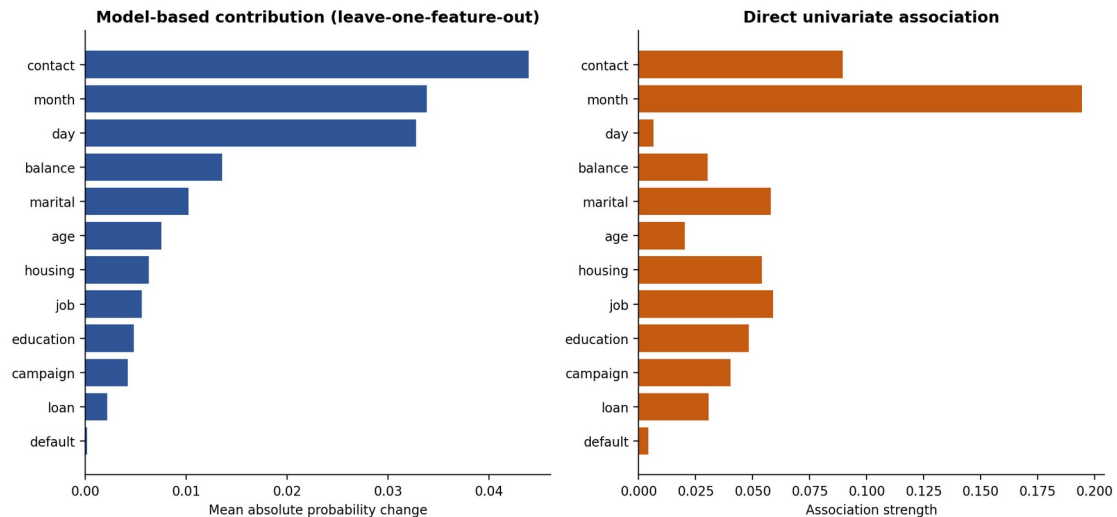


Figure 10 compares model-based feature importance and places month, contact, day, housing and age at the top of the HGB reduction order. day has little direct association but becomes useful with month, illustrating why univariate association and model importance need not agree.

Figure 11. HGB partial-dependence profiles for leading predictors.

How the tuned HGB model uses the five leading pre-call features

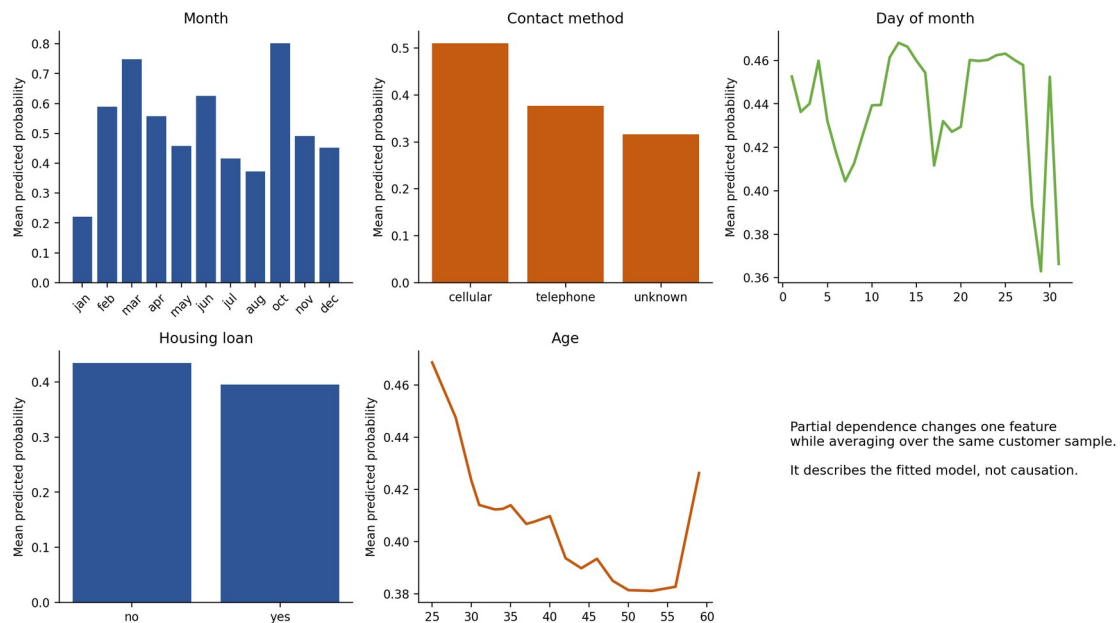


Figure 11 shows the HGB partial-dependence profiles. Partial dependence changes one feature at a time and averages the prediction over the same customer sample, so it describes how the fitted model uses a variable rather than a causal effect. month is strongly non-linear, whereas day is comparatively flat by itself and contributes mainly through interactions.

7.2 Hyperparameter transfer

The reduced models were not retuned. The hyperparameters selected on all pre-call features were transferred unchanged to Top-3, Top-4 and Top-5 sets, so Table 6 isolates the effect of removing features rather than mixing feature reduction with a second tuning exercise.

Table 6. Performance after transferring tuned settings to reduced feature sets.

Model	Feature set	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
LR	Top 3	0.928	0.497	0.026	0.049	0.667	0.195
LR	Top 4	0.928	0.531	0.036	0.067	0.673	0.212
LR	Top 5	0.928	0.514	0.032	0.061	0.674	0.210
LR	All 12	0.928	0.525	0.033	0.062	0.685	0.215
HGB	Top 3	0.928	0.562	0.056	0.102	0.704	0.244
HGB	Top 4	0.928	0.545	0.056	0.101	0.706	0.246
HGB	Top 5	0.928	0.502	0.036	0.066	0.707	0.245
HGB	All 12	0.928	0.509	0.020	0.039	0.715	0.245

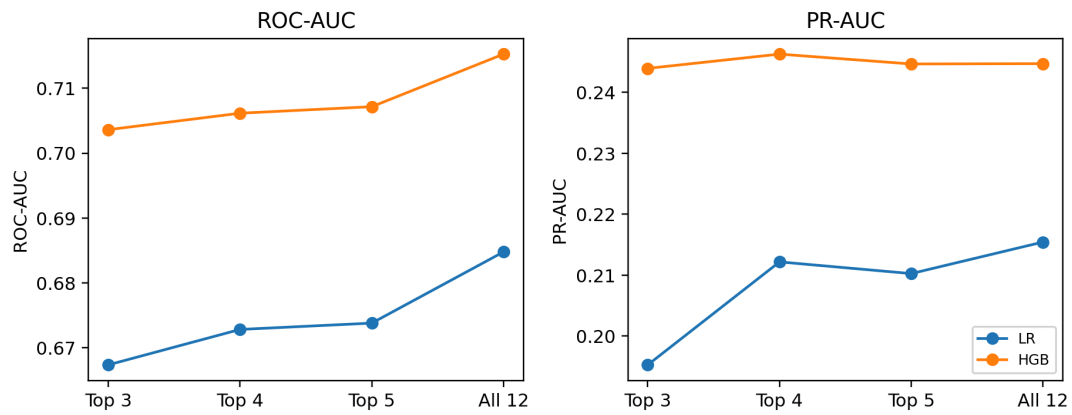


Figure 12 shows that the reduced models remain close under shuffled validation. All 12 features retain the best ROC-AUC for both model families, while some reduced HGB sets have very similar PR-AUC. The combined evidence does not justify dropping variables, and the final model therefore retains the complete 12-feature pre-call set.

8. Class Imbalance

The project compared six approaches from the same tuned-unweighted structural reference. Weighting and resampling are intervention sensitivities rather than separately retuned contests. Threshold tuning uses nested inner out-of-fold predictions inside each outer fold before the final outer-fold model is evaluated.

- **No adjustment:** original data and threshold 0.5.
- **Class weighting:** makes subscriber errors more costly during fitting.
- **Random oversampling:** duplicates subscriber rows in each training fold.
- **Random undersampling:** removes non-subscriber rows in each training fold.
- **SMOTENC:** creates synthetic minority examples while treating categorical variables as categories, not continuous numbers.
- **Threshold optimisation:** leaves training unchanged and selects a lower threshold on an inner validation split to improve F1.

Model	Strategy	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
HGB	None	0.928	0.509	0.020	0.039	0.715	0.245
HGB	Class weight	0.745	0.152	0.547	0.237	0.717	0.247
HGB	Oversampling	0.751	0.156	0.552	0.243	0.719	0.251
HGB	Undersampling	0.723	0.145	0.575	0.231	0.711	0.234
HGB	SMOTENC	0.920	0.398	0.199	0.265	0.684	0.215
HGB	Threshold tuning	0.918	0.398	0.270	0.322	0.715	0.245
LR	None	0.928	0.525	0.033	0.062	0.685	0.215
LR	Class weight	0.661	0.124	0.605	0.205	0.688	0.210
LR	Oversampling	0.662	0.124	0.607	0.206	0.688	0.209
LR	Undersampling	0.659	0.122	0.602	0.203	0.686	0.207
LR	SMOTENC	0.680	0.115	0.513	0.188	0.649	0.164
LR	Threshold tuning	0.900	0.289	0.261	0.274	0.685	0.215

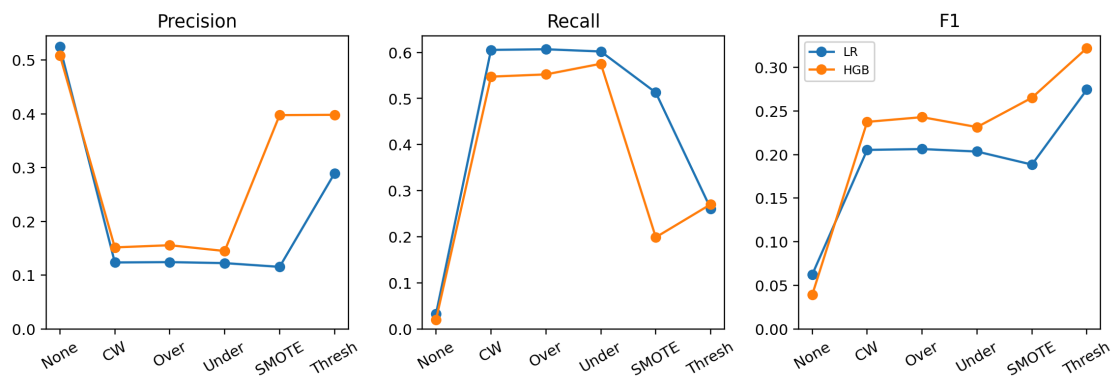


Figure 14. Confusion matrices across imbalance strategies and model families.

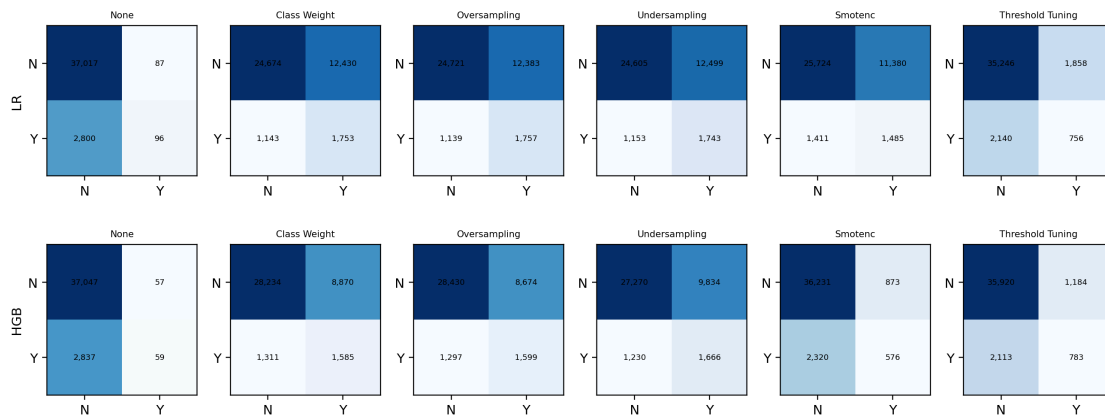


Table 7 and Figures 13-14 show why the intervention matters:

- Logistic Regression responds similarly to weighting, oversampling and undersampling: recall rises sharply while precision and accuracy fall, with little ranking gain.
- HGB is more sensitive to the altered training distribution. Oversampling produces only a small ROC-AUC/PR-AUC increase while substantially reducing precision and accuracy.
- SMOTENC reduces ranking performance for both model families.
- Nested threshold optimisation gives the best F1 without changing the underlying ROC-AUC/PR-AUC because it changes the decision cutoff rather than the score ordering.

For the ranking pilot, the unweighted model is retained because the small resampling gains do not outweigh the operational distortion and added complexity.

Figure 15. Row-stratified versus pseudo-group ROC-AUC sensitivity.

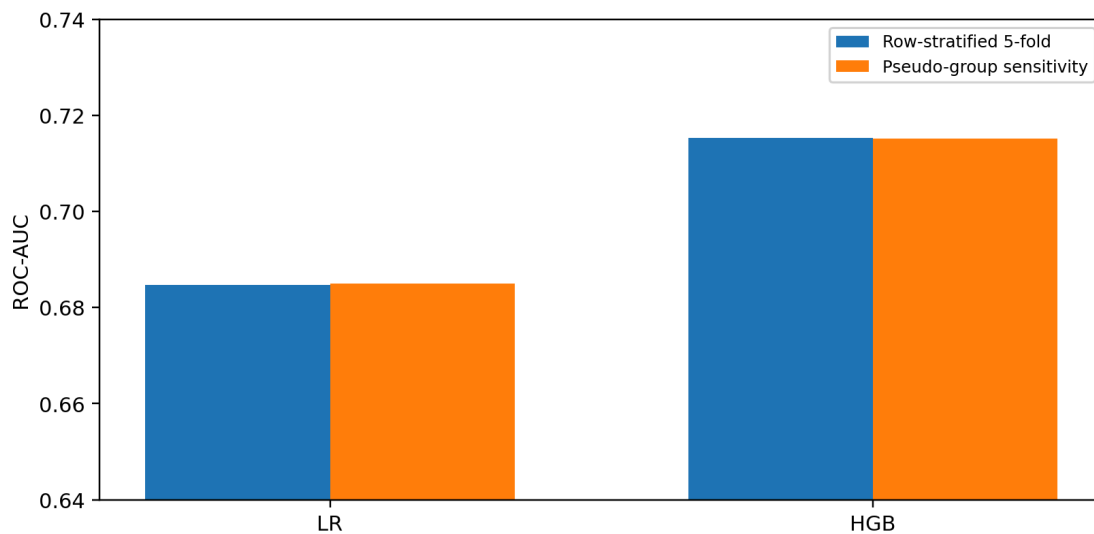


Figure 15 adds a dependence sensitivity check using a pseudo-profile key (age + job + marital + education + balance) because the dataset has no true customer identifier. ROC-AUC remains essentially unchanged (LR about 0.685; HGB about 0.715). This is reassuring, but it is not proof of customer-level independence because the grouping key is only a proxy.

9. Target Customer Population Analysis

9.1 How the population was identified

EDA suggested that customers were more responsive when they were reached by cellular phone and were retired, aged 71 or older, or in the top balance quartile. The exploratory rule was frozen as:

`contact = cellular AND (job = retired OR age >= 71 OR balance >= Q3)`

This produced 6,774 customers with 778 subscriptions: an 11.49% subscription rate versus 7.24% overall.

Two different questions were tested:

1. Does a model trained on the full population perform differently when evaluated on this slice?
2. Does training a dedicated model only on the target population improve performance?

The second analysis is labelled exploratory because the segment rule was discovered from the same dataset.

9.2 Dedicated target-population models

Model	Stage	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
LR	Baseline	0.884	0.420	0.027	0.051	0.693	0.286
LR	Tuned transfer	0.884	0.458	0.035	0.065	0.693	0.286
HGB	Baseline	0.885	0.498	0.130	0.206	0.694	0.303
HGB	Tuned transfer	0.885	0.455	0.026	0.049	0.715	0.309

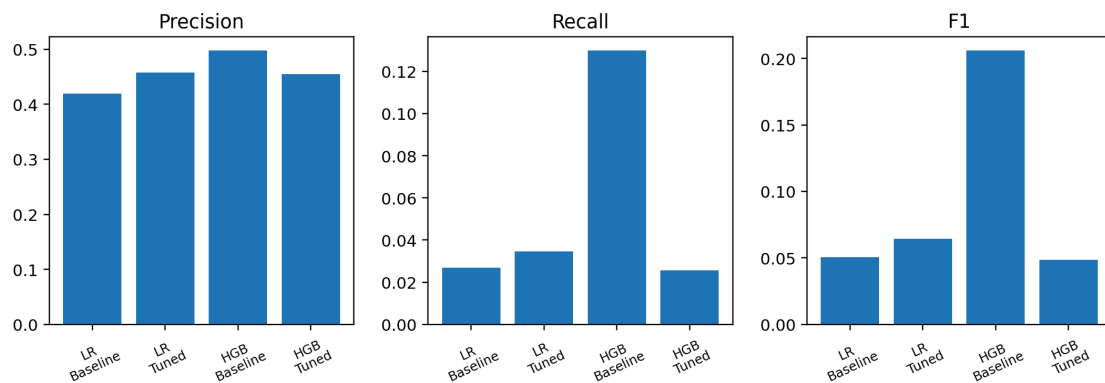


Figure 17. Target-population confusion matrices for baseline and tuned-transfer models.

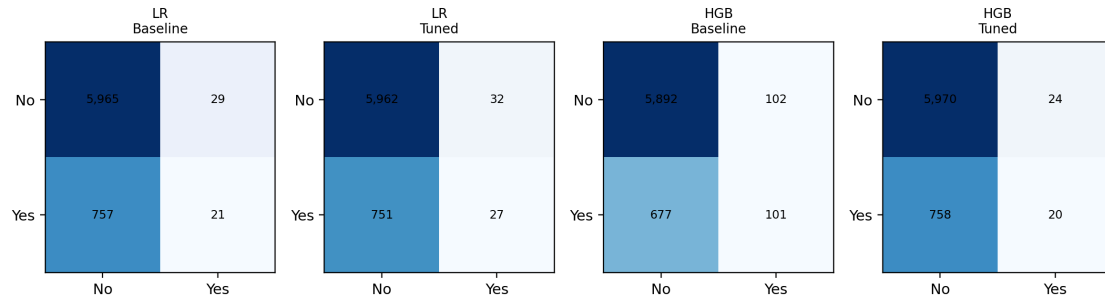


Table 8 summarises the target-population metrics, Figure 16 visualises the precision-recall trade-off, and Figure 17 shows the confusion matrices. Transferred tuning leaves LR almost unchanged, while HGB improves ranking (ROC-AUC about 0.715 versus 0.694) but remains conservative at the default 0.5 threshold. The dedicated segment analysis therefore does not by itself justify a separate production model.

9.3 Transfer policy and exploratory status

The segment rule is exploratory and was discovered from this dataset, so the corrected workflow does not add a second round of segment-specific hyperparameter or imbalance tuning. Instead, the full-population selected settings are transferred unchanged. The HGB transfer improves segment ranking (ROC-AUC about 0.715 versus 0.694 for its baseline) but remains weak at the default 0.5 threshold. The later segment-versus-remainder bootstrap test is therefore the more important robustness check.

9.4 Operating errors and subgroup robustness

A model can perform acceptably on average while missing particular kinds of subscribers. To examine this, the recommended Logistic Regression ranking was converted into a 20% call-depth policy and missed subscribers were grouped by customer and campaign categories.

Figure 18. Concentration of missed subscribers at 20% call depth.

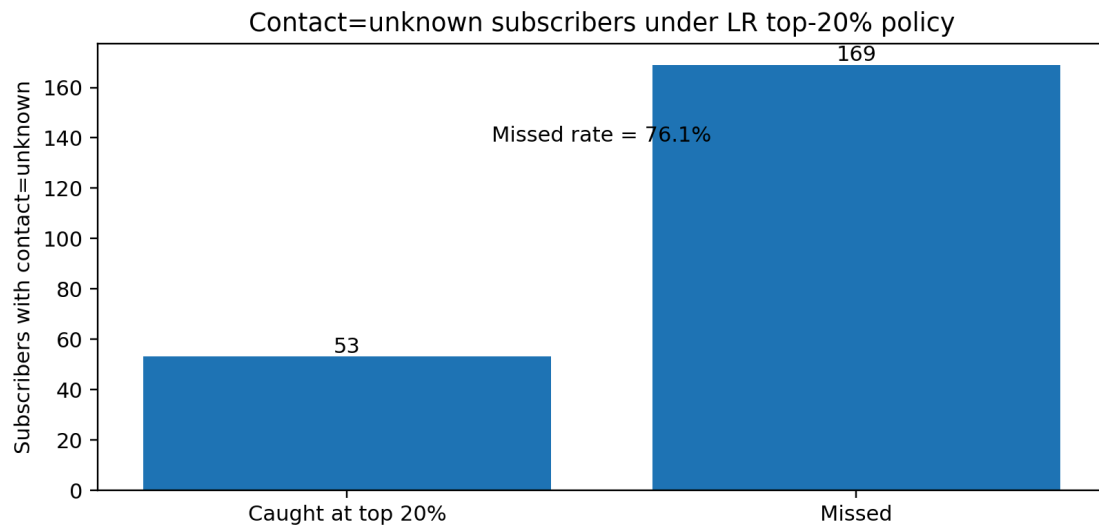


Figure 18 isolates the main operational blind spot in the final forward evaluation. Among subscribers with contact=unknown, 53 were captured and 169 were missed by the Logistic Regression top-20% period-local policy: a missed-subscriber rate of 76.1%. This does not mean the customers were unreachable; it means the recorded contact channel is incomplete and should be audited before live scoring.

The contact=unknown audit is deliberately reported as an operational data-quality issue rather than a reason to remove those customers from the ranked population.

Figure 19. Forward segment-minus-remainder ROC-AUC with 95% bootstrap CIs.

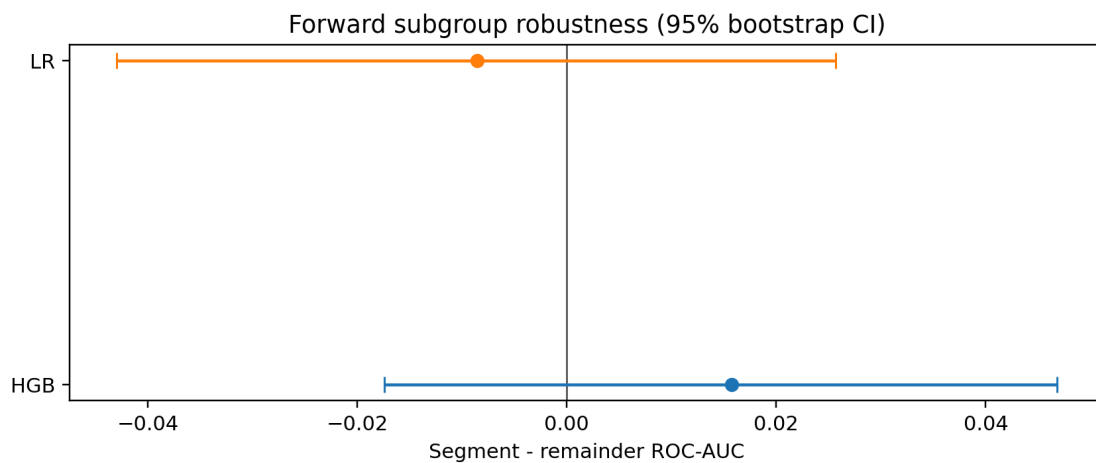


Figure 19 shows the forward segment-minus-remainder ROC-AUC difference. HGB is about +0.016 and LR about -0.009, but both 95% bootstrap intervals cross zero. The segment therefore remains useful as a reporting stratum and hypothesis for a live pilot, not as evidence for a separate production model.

10. Robustness Checks: Rolling Validation, Calibration and Lift

These checks move beyond shuffled development validation. Each forward test block must have at least 5,000 prior training rows and at least 100 positive and 100 negative outcomes; the primary summary is weighted by test-block size.

10.1 What is an expanding or rolling window?

The model trains on the earliest eligible month-labelled blocks and predicts the next block, then expands the training window. Eight blocks satisfy the evidence gate. The ordered blocks are a robustness proxy because the dataset has no verified year or campaign identifier.

Table 9 summarises the result: HGB remains stronger under shuffled development validation, while Logistic Regression has the higher weighted ordered ROC-AUC across later blocks.

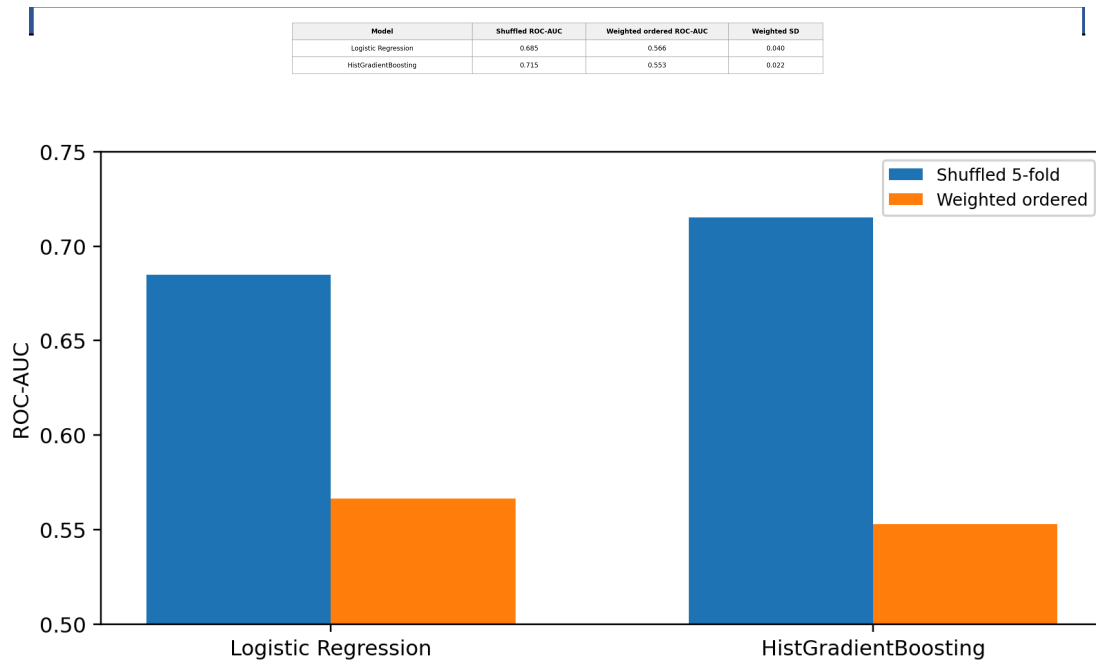


Figure 20 shows the same reversal visually. The weighted ordered ROC-AUC is about 0.566 for Logistic Regression and 0.553 for HGB, supporting LR as the safer pilot ranking model even though HGB leads in shuffled development validation.

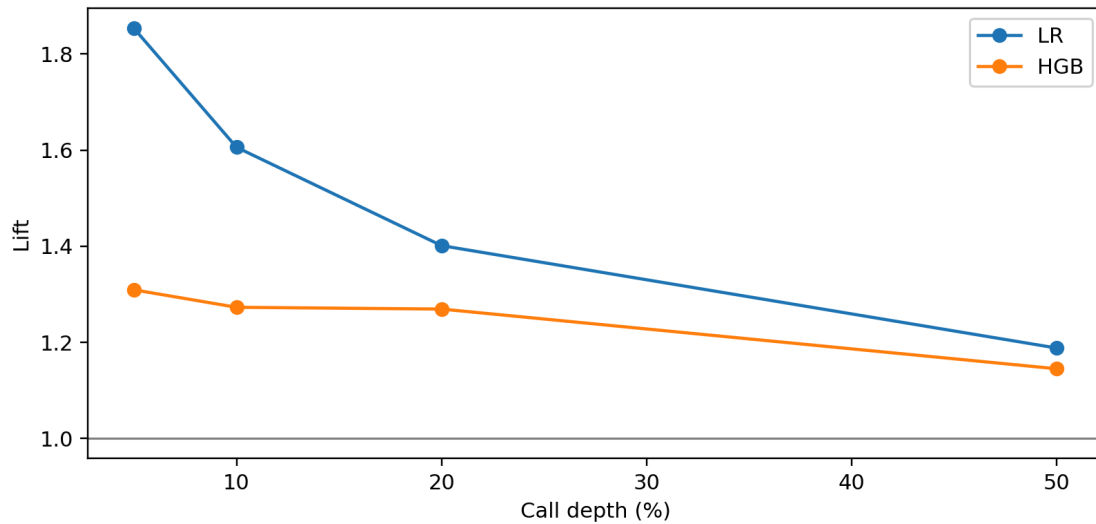
10.2 Lift

Lift asks how much better the ranked call list is than selecting the same number of customers at random. A lift of 1.4 at 20% depth means 40% more subscribers are captured than random selection would be expected to capture. Figure 21 visualises how lift changes as call depth expands.

Customers were ranked within each campaign period because a real campaign ranks only the customers available at that time. Table 10 gives the exact period-local subscriber captures, random expectations and resulting lift values.

Model	Call depth	Subscribers captured	Random expectation	Incremental	Lift
Logistic Regression	5%	228	123	+105	1.85
Logistic Regression	10%	395	246	+149	1.61
Logistic Regression	20%	689	492	+197	1.40
HistGradientBoosting	5%	161	123	+38	1.31
HistGradientBoosting	10%	313	246	+67	1.27
HistGradientBoosting	20%	624	492	+132	1.27

Figure 21. Period-local lift as call depth expands.



10.3 Calibration

Calibration asks whether the score level matches the observed subscription rate. The corrected workflow assesses calibration within each eligible forward period before aggregating, rather than pooling heterogeneous model outputs as though they came from one fitted model.

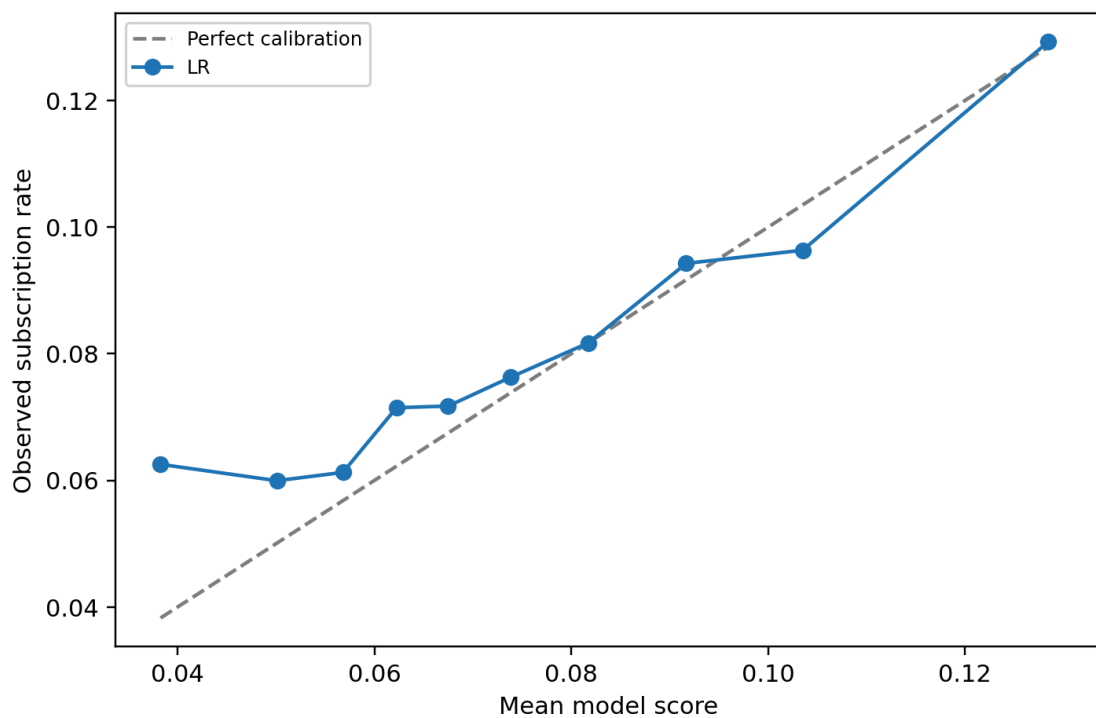


Figure 22 shows that the Logistic Regression ranking is reasonably ordered but not stable enough to treat raw scores as literal probabilities. Across the eight eligible periods, the mean LR score is 7.54% versus an observed 8.05% subscription rate, with period-weighted ECE about 0.044. Prospective recalibration is still required before probability-based financial calculations.

11. Final Comparison and Recommendations

11.1 Model leaderboard

Evidence question	Best result	Interpretation
Best shuffled pre-call ROC-AUC	HGB, 0.715	Best development ranking
Best shuffled F1 after threshold selection	HGB, 0.322	Best hard-classification F1
Best weighted ordered ROC-AUC	LR, 0.566	More reliable ordered ranking
Best period-local lift at 20%	LR, 1.40	More incremental subscribers
Best reduced feature set	All 12 pre-call features	Reduced sets do not justify dropping signal
Best sampling policy	None for pilot	Resampling changes trade-offs; unweighted model is simpler
Best target-population model	No clear separate winner	Bootstrap intervals include zero

Table 11. Final evidence leaderboard supporting model selection.

11.2 Final model decision

Table 11 summarises the synchronized evidence. HGB is the strongest shuffled development ranker, while unweighted Logistic Regression with all 12 pre-call features is preferred for the pilot because it has the stronger weighted ordered ROC-AUC and period-local lift.

- transfers better to future periods;
- produces stronger period-local lift;
- is easier to explain and monitor;
- avoids the probability distortion caused by class weighting;
- does not depend on synthetic observations.

The model should rank customers rather than automatically approve, reject or exclude them.

11.3 Business recommendations

1. Exclude **duration** from all pre-call targeting.
2. Use all 12 available pre-call features; do not reduce to top 3-5 for the pilot.
3. Use unweighted Logistic Regression to rank customers within each campaign period.
4. Select call depth from operational capacity rather than the default 0.5 threshold.
5. Audit the meaning and pre-call availability of **contact=unknown**.
6. Run a controlled pilot: randomly compare model-ranked calling with current practice at the same call volume.
7. Measure subscriptions per 1,000 calls, cost per acquisition, complaints and customer mix.
8. Recalibrate probabilities before using them for revenue or expected-value calculations.

Figure 23. Controlled pilot and feedback loop for prospective deployment.

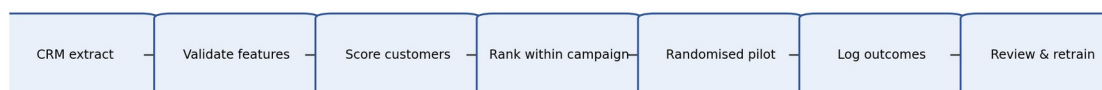


Figure 23 translates the recommendation into a controlled pilot and feedback loop. The pilot is a limited, monitored test designed to determine whether model-ranked calling creates real incremental value under current campaign conditions. The model-ranked group and current-practice control must have the same call volume and run in the same period. The primary measure should be subscriptions per 1,000 calls, supported by cost per acquisition, complaints, opt-outs and customer-mix checks. Scale-up should occur only if uplift is positive, operationally meaningful and not accompanied by unacceptable group-level disparities. A realised lift below 1.0 should trigger immediate suspension and a return to the existing process.